When Do Photos Prove a Blade Is Damaged?

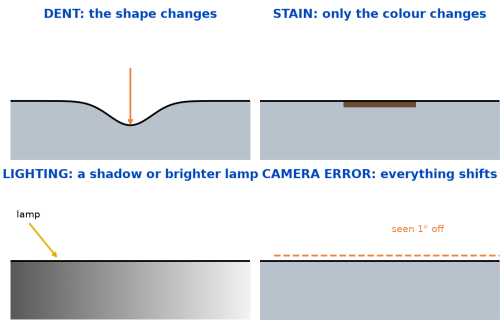
Finding real 3D damage on turbine blades from a CAD model and a few photos, and knowing when the photos are not enough

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1. The problem

Engine blades are checked using photos. But a dark spot in a photo can have four causes: a **dent** (real damage), a **stain**, a **change in lighting**, or the **camera being slightly off**. In a photo they can look almost the same. A 0.1 mm dent changes the image only a tiny bit, while a 1° camera error changes thousands of pixels.

An inspector needs to know: *is there real damage, where is it, and how big is it?* Just as important, a computer should **not invent damage** that the photos cannot actually show.



The same spot can come from different causes.

2. What is missing in current research

- **Anomaly detection methods** (e.g. PAD 2023, SplatPose 2024, PIAD 2025) mark *unusual pixels* in a photo. They do not say how deep or large the damage is in 3D.
- **CAD-based 3D reconstruction** (e.g. CAD-NeRF 2024, CADSplat 2026) builds a 3D copy of an object. Differences from the CAD are treated as noise to absorb, not as damage to measure.
- **3D anomaly datasets** (MVTec 3D-AD, Real3D-AD, SiM3D) give anomaly scores, not measured damage size.

The gap: no method measures real 3D damage *and* says honestly when the photos cannot support a conclusion. An author of PIAD, contacted about this work, independently suggested this exact question as a meaningful research contribution.

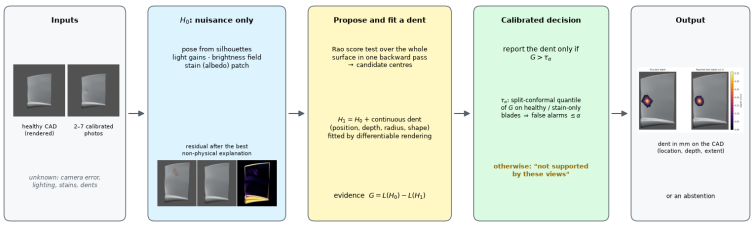
3. Research question

Under a controlled camera and lighting setup, when can photos prove that a blade is physically damaged, rather than stained, lit differently or photographed slightly off? And when should the method say "not enough evidence"?

4. Our approach

Main idea: render and compare. We know exactly what the healthy blade looks like (its CAD model). We use a renderer to draw the blade on the computer and compare it with the real photos.

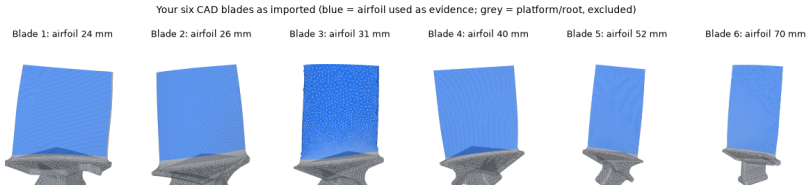
- **Input:** the healthy CAD model + 2–7 photos from known camera positions.
- **Step 1, explain without damage:** adjust camera, lighting and possible stains so the drawing matches the photos as well as possible.
- **Step 2, explain with damage:** do the same, but also allow a dent somewhere on the blade.
- **Step 3, decide:** if adding a dent makes the match *clearly* better, report the dent with its 3D location and depth in mm. If not, say "not enough evidence". The bar for "clearly better" is set using healthy and stained blades, so false alarms stay low (about 5%).
- **Bonus:** the method can predict in advance which dents the cameras are able to see.



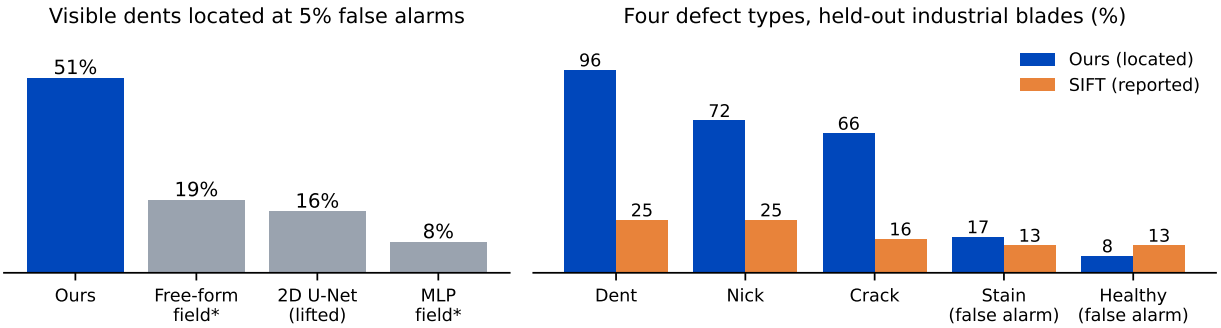
From left to right: CAD + photos → explain without damage → try a dent → decide → 3D dent or "not enough evidence".

5. Results so far (on computer-rendered blades)

We built a test set of 1,152 cases: public Rotor37 blades plus six industrial blade designs. The test blades were kept hidden until the method was final.



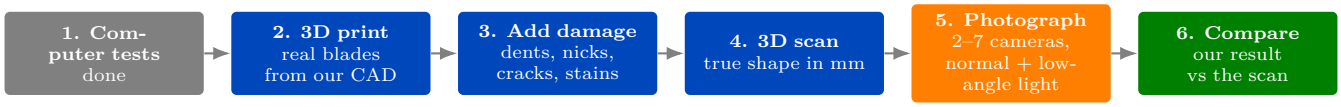
The six industrial blade designs used for testing.



Left: compared with other methods at the same false-alarm rate (*our versions of the CADSplat idea). Right: four damage types; stains and healthy blades should NOT be reported.

- **Finds 51%** of visible dents at 5% false alarms; other methods find at most 19%. Almost all dents deeper than 0.2 mm are found (12 of 13).
- **Works on new blade designs:** after one fix, false alarms dropped from 51% to 4%.
- **Finds nicks and cracks too** (72% and 66%), although it was only designed for dents. A standard image-matching method (SIFT) finds only 16–25%.
- **Standard statistics are not enough:** textbook tests produce 32% and 96% false alarms instead of the intended 1%; setting the bar on real examples is necessary.
- **Main limitation:** the method finds *where* damage is very well, but on the industrial blades it measures dents about 40% too shallow. Adding more cameras finds more dents but does not fix depth, so better lighting is the next thing to try.

6. Plan: from computer blades to real blades



We measure the same things as before: is the damage found, is it in the right place, how accurate is the depth, and how often healthy or stained blades are wrongly flagged.

7. Next experiments

- **Low-angle (grazing) light:** dents cast small shadows, which should make depth measurable.
- **Retrain the appearance model** on industrial blades, to fix the depth underestimate.
- **Re-set the decision bar** on industrial blades, to bring false alarms back to 5%.
- **More camera views** aimed at a suspicious spot.
- **Real photos:** align real photos with the CAD (markers or SIFT) and test on a public real dataset (SiM3D).

8. Expected contributions

1. A clear way to ask *when photos can prove damage*, with a method that says “not enough evidence” when they cannot.
2. A working method that finds and measures 3D damage with controlled false alarms.
3. Evidence that *where* damage is can be found reliably, while *how deep* it is needs better lighting.
4. A test set of computer-rendered and real, scanned blades for this problem.

9. What we need

3D printer, 3D scanner, a calibrated multi-camera setup (2–7 views), controllable lights including one low-angle light, and GPU time. **Main risk:** real metal reflects light differently from the computer model; we will start with matte printed blades and set the decision bar using real healthy blades.